



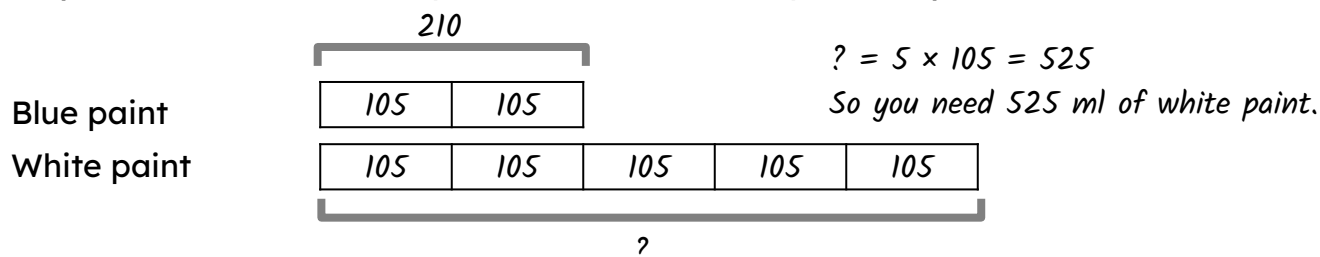
# Problem solving with fractions and ratios

## Task A: Strategies to solve ratio problems

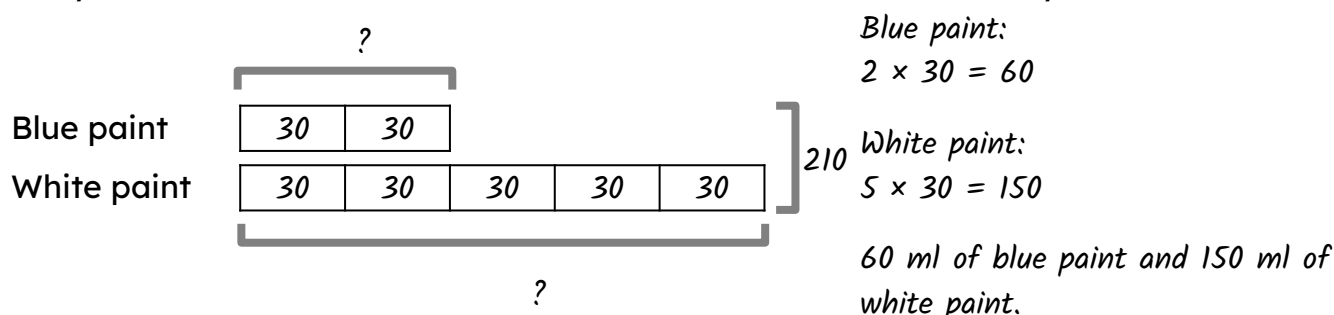
1) 1) Solve these problems. You may want to draw a bar model to help you.

To make the required shade of paint you need to mix blue and white paint in the ratio of 2 : 5

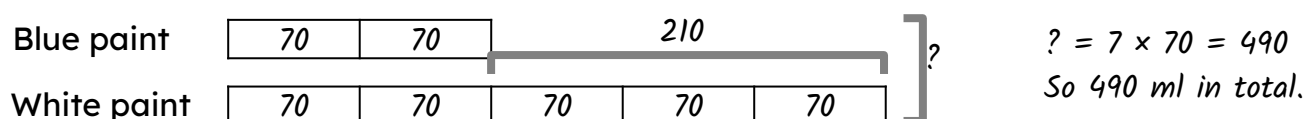
a) If you have 210 ml of blue paint, how much white paint do you need?



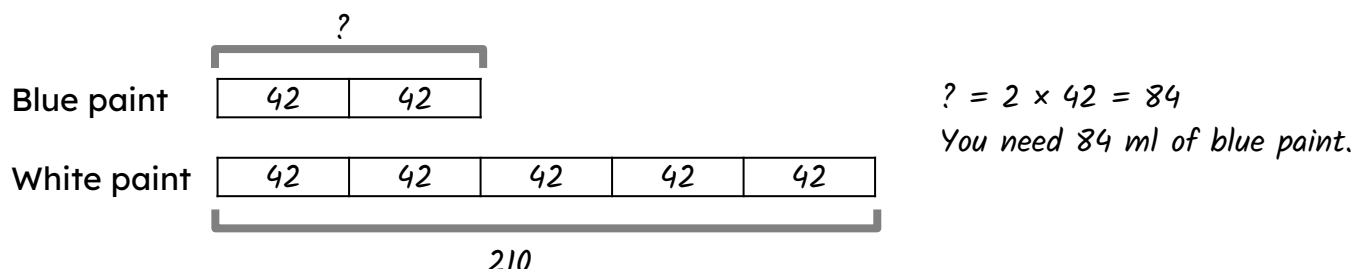
b) If you need to make 210 ml in total, how much of each colour do you need?



c) If you used 210 ml more white paint than blue paint. How much paint was made in total?



d) If you have 210 ml of white paint, how much blue paint do you need?



**Task B: Problems involving fractions and ratios**

- 1) A bag of sweets contains red, green and yellow sweets.

$\frac{1}{3}$  of the bag are red.

The rest of the sweets are either green or yellow, in the ratio 3 : 5

There are 72 sweets in the bag.

How many of each colour are there?

$$\left. \begin{array}{l} \text{Red} = 24 \\ \text{Green} = 18 \\ \text{Yellow} = 30 \end{array} \right\} \text{Check } 24 + 18 + 30 = 72$$

- 2) Izzy counts 350 red, blue and white cars in a car park.

The ratio of red to blue cars is 2 : 3

$\frac{4}{7}$  of the cars are white.

How many of each colour are there in the car park?

$$\text{White} = 200$$

$$\text{Red} = 60$$

$$\text{Blue} = 90$$

- 3) Andeep's older sister Belle is 5 times his age.

The cat is three-tenths of Belle's age.

The total of Andeep's and Belle's ages is 24.

What is the total of their ages?

$$\left. \begin{array}{l} \text{Andeep} = 4 \\ \text{Belle} = 20 \\ \text{Cat} = 6 \end{array} \right\} \text{Total age} = 30$$

